

Voice of Northern: A Digital Platform for Student Rights, Engagement, and Community Empowerment

Abstract:

University students often face challenges such as lack of a structured platform to raise their voices, report injustices, access reliable information, and actively participate in community-driven initiatives. Most student organizations rely heavily on fragmented communication channels like social media and messaging groups, which can lead to information loss, limited transparency, inefficient complaint handling, and reduced organizational scalability. These limitations make it difficult to systematically document student issues, coordinate events, and ensure timely action toward protecting student rights.

The **Voice of Northern** project aims to address these challenges by developing a centralized, modern, and secure digital platform tailored for a university student organization. The platform is designed to provide an integrated environment where students can easily submit complaints (with optional anonymity), track issue resolution, access official notices, participate in events, read publications, and engage with the student community. By digitizing organizational workflows, the system improves transparency, accountability, and accessibility while strengthening communication between students and the organization.

The expected outcome of this project is a full-featured, production-level web platform that empowers students through structured voice representation, efficient complaint management, and active community participation. The system will enhance organizational efficiency, promote student rights advocacy, and create a sustainable digital ecosystem for student engagement, leadership, and social impact within the university.

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Requirement Analysis

Project Title

Voice of Northern: Digital Platform for Student Rights, Engagement, and Organizational Management

1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed requirement analysis of the *Voice of Northern* digital platform. It describes system objectives, stakeholders, functional and non-functional requirements, constraints, and requirement models to guide the design and development of the system.

1.2 Scope

Voice of Northern is a centralized web-based platform designed for a university student organization to manage student rights advocacy, complaints, events, announcements, publications, and member activities. The system aims to replace fragmented communication channels (e.g., Facebook, WhatsApp) with a structured, secure, and scalable digital environment.

1.3 Definitions

- **Complaint** — A report submitted by a student regarding injustice, problem, or issue.
 - **Anonymous Complaint** — Complaint without revealing identity.
 - **Admin** — Full system controller.
 - **Moderator** — Operational manager.
 - **Member** — Registered student user.
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2. Overall Description

2.1 Product Perspective

The system is a **web-based centralized platform** consisting of:

- Public website
- Complaint management system
- Event management system
- Membership management
- Content & notice system
- Admin dashboard

It integrates organizational workflow digitally.

2.2 Stakeholders

Stakeholder	Interest
Students	Submit complaints, participate in events
Organization Admin	Manage system & decisions
Moderators	Handle complaints/events
University Community	Access information
Developers	Maintain system

2.3 Problem Statement (From Requirements Understanding)

Current Issues:

- No structured complaint system

- Fragmented communication (Facebook, Messenger, WhatsApp)
 - Lack of transparency and tracking
 - Poor documentation of issues
 - Inefficient event & member management
 - Limited scalability
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2.4 Product Objectives

The system must:

- Provide centralized platform
 - Enable complaint submission & tracking
 - Improve transparency & communication
 - Manage events and notices
 - Maintain member database
 - Provide admin control
 - Ensure privacy and security
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3. Functional Requirements (System Behavior)

3.1 User Management

- User can register as member
- Admin can assign roles (Admin/Moderator/Member)
- User profile management

- Login / Authentication
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3.2 Complaint Management (Core Feature)

System shall allow:

- Submit complaint
 - Submit anonymously
 - Upload evidence
 - Track complaint status
 - Admin assign complaint
 - Update complaint progress
 - Close / resolve complaint
 - Maintain complaint history
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3.3 Event Management

System shall:

- Create event
- Publish event
- Register participants
- Track participants
- Manage event records
- Publish results / updates

3.4 Notice & Announcement

System shall:

- Publish notices
- Categorize notices
- Show urgent notices
- Display notice archive

3.5 Blog / Magazine

System shall:

- Publish articles
- Upload magazine PDF
- Categorize posts
- Allow reading & browsing

3.6 Membership System

System shall:

- Allow membership application
- Maintain member database
- Manage volunteer list
- Store department/batch/skills

3.7 Admin Panel

Admin shall:

- View dashboard analytics
- Manage complaints
- Manage events
- Publish notices
- Manage users
- Assign roles
- Monitor activity

3.8 Notification System

System shall:

- Notify complaint updates
- Send event reminders
- Broadcast announcements

4. Non-Functional Requirements

4.1 Performance

- Fast response time (<2 sec typical)
- Support multiple concurrent users

- Optimized complaint submission (<30 sec)
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4.2 Security

- Role-based access control
 - Secure authentication
 - Anonymous complaint protection
 - Data encryption
 - File validation
 - Backup & recovery
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4.3 Usability

- Simple interface
 - Mobile responsive
 - Easy navigation
 - Accessible design
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4.4 Reliability

- 24/7 availability
 - Error handling
 - Data integrity
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4.5 Scalability

- Support increasing users
 - Modular architecture
 - Extendable features
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4.6 Maintainability

- Clean architecture
 - Modular code
 - Documentation
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5. Requirements Modeling (Pressman Ch-8)

5.1 Scenario-Based Model (Use Cases)

Primary Use Cases:

- Submit Complaint
 - Track Complaint
 - Register for Event
 - Publish Notice
 - Manage Complaint
 - Manage Users
 - Submit Membership
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5.2 Data Model (Conceptual Entities)

Main Entities:

- User
- Complaint
- ComplaintLog
- Event
- Registration
- Notice
- Blog
- Media

Relationships:

- User submits Complaint
- Admin manages Complaint
- User registers Event
- Admin publishes Notice

5.3 Functional Model (Major System Functions)

- Authentication
- Complaint workflow
- Event workflow
- Notification

- Content publishing
 - Role management
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5.4 Behavioral Model

Complaint Lifecycle:

Submit → Review → Assigned → Processing → Resolved

Event Lifecycle:

Created → Published → Registration → Completed → Archived

6. External Interface Requirements

6.1 User Interface

- Web browser interface
- Mobile responsive UI
- Dark mode support

6.2 Hardware Interface

- Server / VPS
- Cloud storage

6.3 Software Interface

- Database (PostgreSQL)
- Email service
- Cloud storage (optional)

7. Constraints

- Must protect anonymous complaints
 - Limited budget possible
 - Internet dependency
 - University policy compliance
 - Data privacy required
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8. Assumptions

- Students have internet access
 - Admin team will manage system
 - Complaint data handled responsibly
 - System will grow gradually
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9. Requirement Validation

Requirements are valid if system:

- Allows complaint submission & tracking
- Supports event & notice management
- Maintains member database
- Provides admin control

- Ensures security & privacy
 - Improves communication efficiency
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10. Expected Outcome

The system will:

- Centralize organizational workflow
- Improve transparency & accountability
- Empower student voice
- Digitize complaint handling
- Improve engagement & participation
- Provide scalable digital platform